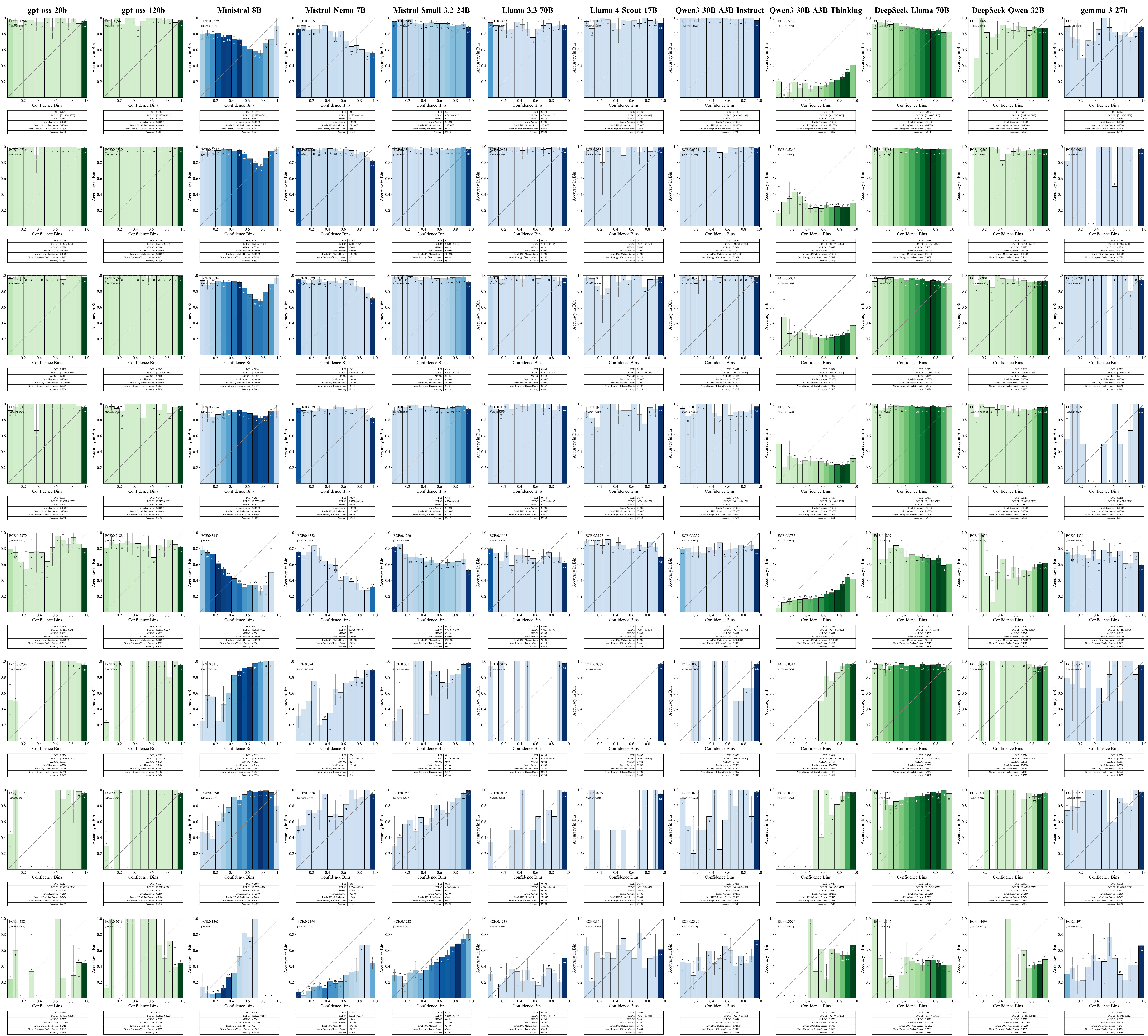
Calibration Plots for Uncertainty Metric P(True)



MMLU

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.1254	0.0963	0.3379	0.6015	0.3545	0.3433	0.0830	0.1137	0.5266	0.2382	0.0683	0.1170
ECE CI	[0.1185–0.1323]	[0.0907–0.1022]	[0.3287–0.3478]	[0.5922–0.6114]	[0.3457–0.3627]	[0.3343–0.3527]	[0.0768–0.0892]	[0.1078–0.1199]	[0.5177–0.5357]	[0.2308–0.2463]	[0.0616–0.0748]	[0.1100–0.1238]
AUROC	0.4892	0.5337	0.3985	0.3151	0.3837	0.4510	0.4459	0.4182	0.6177	0.4303	0.5069	0.5483
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	2/10000	0/10000	0/10000	374/10000	121/10000	20/10000	0/10000	375/10000	12/10000	48/10000	64/10000	0/10000
Norm. Entropy of Bucket Counts	0.2670	0.2692	0.9416	0.5994	0.8821	0.4078	0.1968	0.3173	0.7538	0.8833	0.4958	0.1326
Accuracy	0.9578	0.9684	0.7218	0.7458	0.9232	0.9200	0.9360	0.9738	0.2499	0.8651	0.8779	0.8861

ARC-Easy

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0736	0.0730	0.2935	0.5206	0.1311	0.0873	0.0331	0.0354	0.5266	0.3181	0.0565	0.0086
ECE CI	[0.0689–0.0783]	[0.0689–0.0770]	[0.2872–0.3012]	[0.5114–0.5295]	[0.1280–0.1365]	[0.0818–0.0927]	[0.0304–0.0358]	[0.0324–0.0385]	[0.5177–0.5352]	[0.3135–0.3228]	[0.0538–0.0605]	[0.0052–0.0117]
AUROC	0.5780	0.5086	0.3774	0.2644	0.4628	0.4742	0.5636	0.5074	0.4989	0.4606	0.5222	0.5364
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	618/10000	34/10000	3/10000	0/10000	22/10000	2/10000	1/10000	1/10000	0/10000
Norm. Entropy of Bucket Counts	0.1487	0.1631	0.8976	0.6722	0.6685	0.1857	0.0515	0.1061	0.7532	0.8702	0.4666	0.0096
Accuracy	0.9862	0.9936	0.8938	0.9144	0.9752	0.9824	0.9878	0.9844	0.2498	0.9748	0.9653	0.9732

ARC-Challenge

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.1100	0.0847	0.3034	0.5629	0.1802	0.1008	0.0253	0.0397	0.5034	0.2958	0.0801	0.0291
ECE CI	[0.1036–0.1169]	[0.0801–0.0894]	[0.2969–0.3125]	[0.5540–0.5710]	[0.1748–0.1854]	[0.0957–0.1077]	[0.0211–0.0293]	[0.0353–0.0444]	[0.4946–0.5124]	[0.2903–0.3023]	[0.0744–0.0856]	[0.0246–0.0343]
AUROC	0.5117	0.4688	0.3700	0.2657	0.3686	0.4633	0.5156	0.4494	0.5392	0.4244	0.4972	0.5037
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	242/10000	0/10000	0/10000	520/10000	42/10000	2/10000	0/10000	8/10000	1/10000	0/10000	1/10000	0/10000
Norm. Entropy of Bucket Counts	0.2209	0.2021	0.9139	0.6418	0.7321	0.2367	0.0827	0.1266	0.7819	0.8787	0.5730	0.0266
Accuracy	0.9770	0.9878	0.8405	0.8528	0.9514	0.9691	0.9713	0.9729	0.2500	0.9394	0.9277	0.9445

SciQ

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0317	0.0475	0.2654	0.4829	0.1826	0.0850	0.0237	0.0137	0.5186	0.3168	0.0717	0.0168
ECE CI	[0.0292–0.0372]	[0.0430–0.0523]	[0.2579–0.2731]	[0.4738–0.4920]	[0.1766–0.1885]	[0.0798–0.0907]	[0.0201–0.0273]	[0.0112–0.0178]	[0.5103–0.5281]	[0.3125–0.3218]	[0.0668–0.0760]	[0.0127–0.0210]
AUROC	0.5032	0.4960	0.4617	0.3470	0.4426	0.5181	0.5819	0.5319	0.5014	0.4855	0.5392	0.5364
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	1/10000	0/10000	0/10000	927/10000	32/10000	4/10000	0/10000	0/10000	0/10000	2/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.1027	0.1278	0.8890	0.6650	0.7759	0.2018	0.0698	0.0590	0.7616	0.8641	0.5378	0.0100
Accuracy	0.9658	0.9756	0.8809	0.8968	0.9549	0.9762	0.9752	0.9674	0.2501	0.9686	0.9520	0.9501

GPQA

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.2370	0.2188	0.3133	0.6522	0.4286	0.5007	0.3177	0.3259	0.3735	0.1807	0.3030	0.4339
ECE CI	[0.2283–0.2457]	[0.2103–0.2270]	[0.3039–0.3237]	[0.6428–0.6610]	[0.4179–0.4390]	[0.4925–0.5106]	[0.3086–0.3269]	[0.3161–0.3370]	[0.3648–0.3829]	[0.1675–0.1945]	[0.2942–0.3134]	[0.4240–0.4436]
AUROC	0.4651	0.4813	0.3205	0.2779	0.3598	0.3885	0.3638	0.4937	0.6397	0.4503	0.5242	0.3869
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/5000	0/10000	0/10000
Invalid UQ Method Scores	85/10000	0/10000	1/10000	382/10000	272/10000	90/10000	3/10000	2250/10000	405/10000	40/5000	179/10000	3/10000
Norm. Entropy of Bucket Counts	0.2665	0.2389	0.8901	0.5030	0.8384	0.5228	0.3631	0.5508	0.9212	0.7815	0.4812	0.3489
Accuracy	0.8016	0.8193	0.5123	0.5678	0.6678	0.7019	0.7234	0.7970	0.2526	0.6598	0.5999	0.6304

GSM8K

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0234	0.0183	0.3113	0.0741	0.0311	0.0138	0.0007	0.0070	0.0314	0.3542	0.0524	0.0574
ECE CI	[0.0153–0.0325]	[0.0109–0.0275]	[0.3009–0.3245]	[0.0651–0.0886]	[0.0256–0.0399]	[0.0070–0.0208]	[0.0001–0.0087]	[0.0030–0.0149]	[0.0274–0.0404]	[0.3423–0.3671]	[0.0426–0.0623]	[0.0476–0.0698]
AUROC	0.6091	0.5726	0.7622	0.6346	0.8006	0.5022	0.4942	0.5363	0.5554	0.5029	0.4908	0.6587
Invalid Answers	4/2500	2/2500	8/2500	12/2500	0/2500	0/2500	0/2500	0/2500	134/2500	0/2500	12/2500	0/2500
Invalid UQ Method Scores	1/2496	0/2498	1/2492	33/2488	0/2500	18/2500	0/2500	6/2500	0/2366	2/2500	0/2488	2/2500
Norm. Entropy of Bucket Counts	0.0650	0.0466	0.7550	0.3161	0.3895	0.0218	0.0000	0.0259	0.3473	0.9342	0.4314	0.1278
Accuracy	0.9459	0.9484	0.8976	0.8582	0.9544	0.9734	0.9660	0.9599	0.9611	0.9492	0.9373	0.9432

SVamp

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.0127	0.0124	0.2690	0.0658	0.0521	0.0108	0.0239	0.0205	0.0346	0.2908	0.0437	0.0778
ECE CI	[0.0086–0.0214]	[0.0076–0.0205]	[0.2592–0.2803]	[0.0546–0.0788]	[0.0449–0.0619]	[0.0061–0.0184]	[0.0157–0.0343]	[0.0140–0.0289]	[0.0287–0.0427]	[0.2792–0.3027]	[0.0358–0.0527]	[0.0684–0.0898]
AUROC	0.5688	0.5813	0.8170	0.7146	0.8978	0.6560	0.6012	0.6761	0.6830	0.6725	0.5479	0.7943
Invalid Answers	4/2500	0/2500	54/2500	39/2500	9/2500	8/2500	1/2500	14/2500	55/2500	201/2500	24/2500	1/2500
Invalid UQ Method Scores	0/2496	0/2500	0/2446	81/2461	0/2491	25/2492	0/2499	0/2486	0/2445	2/2299	1/2476	0/2499
Norm. Entropy of Bucket Counts	0.0975	0.0839	0.8565	0.2644	0.4927	0.0553	0.0243	0.0829	0.3573	0.8868	0.3864	0.1935
Accuracy	0.9559	0.9572	0.9309	0.9164	0.9398	0.9603	0.9400	0.9646	0.9644	0.9316	0.9487	0.9560

SciBench

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.4884	0.5018	0.1363	0.2194	0.1258	0.4238	0.3609	0.2390	0.3024	0.3345	0.4493	0.2914
ECE CI	[0.4687–0.5084]	[0.4829–0.5225]	[0.1223–0.1518]	[0.2035–0.2357]	[0.1080–0.1447]	[0.4065–0.4459]	[0.3431–0.3804]	[0.2247–0.2608]	[0.2797–0.3267]	[0.3159–0.3587]	[0.4286–0.4711]	[0.2743–0.3121]
AUROC	0.5797	0.5112	0.7348	0.6860	0.6859	0.5700	0.6061	0.6644	0.5759	0.4926	0.5270	0.6933
Invalid Answers	45/2500	9/2500	55/2500	60/2500	11/2500	14/2500	9/2500	12/2500	743/2500	328/2500	291/2500	6/2500
Invalid UQ Method Scores	8/2455	1/2491	0/2445	129/2440	0/2489	111/2486	0/2491	10/2488	5/1757	83/2172	34/2209	4/2494
Norm. Entropy of Bucket Counts	0.1869	0.1137	0.6387	0.8817	0.9491	0.3425	0.1489	0.5588	0.5310	0.7588	0.4587	0.4206
Accuracy	0.4140	0.4257	0.1820	0.1809	0.5195	0.4644	0.6078	0.6473	0.5919	0.4385	0.4317	0.5486